

Technical Report

Local Business Recovery Support System
Titan Plaza, Bogotá — Systems Analysis & Design 2026-I

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Abstract

This report presents a complete systems engineering study of the Local Business Recovery Support System, developed for the Titan Plaza shopping mall in Bogotá, Colombia. The study was conducted across four structured phases in the post-pandemic context of 2024–2026, a period marked by significant shifts in consumer behavior that have disproportionately affected small independent retailers.

Phase one involved primary data collection through direct observation, customer and employee surveys, and environmental correlation. Findings revealed that while Titan Plaza attracts substantial foot traffic, this traffic does not translate proportionally into sales for small businesses due to a structural informational disconnect between merchants and customers.

Phase two produced a modular three-layer analytical platform designed to centralize business data, automate inventory alerts, generate demand forecasts, and enable data-driven decision-making for store owners.

Phase three refined the architecture through robust engineering principles, applied ISO 9001 and PMBOK quality and risk management frameworks, and established a comprehensive project management plan.

Phase four validated the design through computational simulation across three scenarios — baseline, optimization, and failure — using SimPy and Matplotlib. Results confirmed that predictive and promotional mechanisms more than doubled baseline sales performance, while identifying synchronization reliability as the most critical factor for sustained system operation.

1 Introduction

The post-pandemic recovery of small retail businesses represents one of the most pressing challenges in urban commerce across Latin America. In Bogotá, shopping malls such as Titan Plaza serve as critical economic ecosystems where large chain stores and small independent businesses coexist, yet compete for the same customer base under vastly unequal conditions. While large retailers leverage digital infrastructure, real-time inventory systems, and data-driven marketing strategies, small businesses continue to operate largely on intuition, informal knowledge, and accidental foot traffic.

This project addresses that structural gap through the design and validation of an analytical support platform specifically tailored to the needs and constraints of small businesses within Titan Plaza. The work progressed through four structured phases: systems analysis, system design, robust engineering and project management, and computational simulation. Together, these phases constitute a comprehensive systems engineering approach that moves from real-world observation to validated design.

2 Literature Review

Consumer behavior in physical retail environments has undergone substantial transformation following the COVID-19 pandemic. Research by McKinsey & Company (2023)

identifies several persistent shifts: customers increasingly prioritize value for money, combine online and in-person shopping experiences, and seek experiential retail over transactional product acquisition. These changes have accelerated the competitive disadvantage of small businesses, which lack the digital infrastructure to adapt to evolving customer expectations.

Within the context of shopping malls, foot traffic patterns have become increasingly complex and context-dependent. Studies on retail analytics highlight the critical role of real-time data in enabling merchants to anticipate demand fluctuations, optimize inventory, and engage customers through timely promotions. The absence of such tools creates what systems theorists describe as a fragmented information environment, where individual agents operate without visibility into broader system-level patterns (Sommerville, 2016).

From a systems engineering perspective, small business ecosystems in mixed retail environments exhibit properties of complex adaptive systems, including emergent behavior, feedback loops, and sensitivity to external variables such as weather, day of the week, and macroeconomic conditions. Addressing these characteristics requires architectural designs that account for variability, resilience, and scalability (Pressman & Maxim, 2020).

3 Objectives

The primary objective of this project is to design and validate an analytical platform that supports the post-pandemic recovery of small local businesses in Titan Plaza by reducing the informational disconnect between merchants and customers.

Specific objectives include:

- Identifying the key factors that prevent the conversion of visitor foot traffic into sales for small businesses within the mall.
- Designing a system architecture that centralizes sales, inventory, and customer behavior data into a unified dashboard accessible to store owners.
- Proposing automated mechanisms for inventory alerts and demand forecasting based on observed traffic and purchasing patterns.
- Applying robust engineering principles and risk management frameworks to ensure system reliability, scalability, and quality.
- Validating the proposed design through computational simulation across multiple operational scenarios.
- Defining a deployment and implementation strategy that promotes gradual adoption among store owners.

4 Scope

This project focuses exclusively on the small independent businesses operating within Titan Plaza, located on Av. Boyacá with Calle 80 in Bogotá. The mall operates daily

from 8:00 a.m. to 10:00 p.m. and houses approximately 250 stores across four commercial floors.

Within scope: small independent stores, their employees, store owners and managers, customers who interact with these businesses, and the common areas of the mall insofar as they influence foot traffic patterns.

Outside scope: large chain anchor stores (Falabella, Éxito), the mall’s administrative offices, loading and unloading areas, external suppliers, and any businesses located outside the physical boundaries of the mall.

5 Assumptions

The following assumptions were made throughout the development of this project:

- The foot traffic patterns observed on the two data collection days are sufficiently representative of general weekly behavioral trends, acknowledging the limitations of a small sample.
- Store owners are willing to participate in the platform voluntarily once trust is established through a pilot phase and direct onboarding support.
- The mall’s internet connectivity, while variable in some areas, is sufficient to support the system during peak usage hours with offline synchronization as a fallback.
- Existing third-party inventory systems used by some store owners can be integrated with or coexist alongside the proposed platform without requiring owners to abandon their current tools.
- The behavioral and spending patterns identified in the survey data reflect broader trends in the post-pandemic consumer landscape, consistent with findings from McKinsey & Company (2023).
- Simulation parameters derived from Workshop #1 data are sufficiently representative to produce valid behavioral insights across the evaluated scenarios.

6 Limitations

Several limitations affected the scope and reliability of the findings:

The primary data collection was based on a relatively small survey sample of 17 respondents, which limits the statistical representativeness of the results. Responses may also reflect individual perceptions rather than objective behavioral patterns, introducing potential bias.

Observation sessions were conducted on only two days — one Saturday and one Sunday — which may not capture the full range of weekly or seasonal variability. External factors such as specific promotions, weather conditions, and calendar events during the observation period may have influenced the data in ways that are not generalizable.

The proposed system has not been deployed or validated in a real operational environment. Performance metrics and optimization outcomes observed in the simulation represent

model-based projections rather than empirical measurements from a live system, and simulation parameters are based on the limited primary data available from the field study.

7 Methodology

7.1 Phase 1 — Systems Analysis

The analysis of Titan Plaza was conducted using three complementary data collection methods selected to provide both quantitative and qualitative perspectives on the small business ecosystem.

Direct Observation involved structured monitoring of visitor flow entering and exiting the mall across two sessions: Saturday from 9:00 a.m. to 12:00 p.m., and Sunday from 12:00 p.m. to 3:00 p.m. The observation focused on identifying areas of high visitor concentration, behavioral patterns in different zones, and the relationship between visitor activity and store engagement.

Surveys were administered to 17 participants including both customers and store employees. Questions addressed visit motivations, spending behavior relative to pre-pandemic levels, and product availability experiences in small stores.

Environmental Correlation examined the relationship between external variables — including weather, day of the week, time of day, and in-store promotions — and observed foot traffic levels, enabling the identification of sensitivity factors and variability drivers within the system.

7.2 Phase 2 — System Design

The design phase translated analytical findings into a structured architectural proposal using systems engineering principles. Findings from the analysis were first converted into functional and non-functional requirements, which then guided the definition of system components, data flows, interfaces, and implementation strategies. The system architecture was designed around three layers — input, processing, and output — following a modular structure that separates data collection, business logic, and user-facing reporting.

7.3 Phase 3 — Robust Design and Project Management

The system architecture was refined by applying robust engineering principles including modularity, fault tolerance, scalability, and maintainability, aligned with ISO 9001 quality standards and PMBOK risk management guidelines. A comprehensive risk management plan was developed covering technical, operational, and user-related risks. A quality assurance framework defined acceptance criteria including 99.5% system uptime, response time targets, and a minimum 70% positive usability feedback rate. A project management plan was established covering team roles, implementation phases, resource allocation, and a communication and control strategy.

7.4 Phase 4 — Simulation and Validation

Two complementary simulation approaches were implemented using SimPy, Tkinter, and Matplotlib. A **process-oriented simulation** modeled weekly operational workflows including customer arrivals, sales processing, inventory updates, alert generation, and synchronization activities. A **behavior-oriented simulation** modeled adaptive interactions, feedback loops, and emergent behaviors under varying conditions. Three scenarios were evaluated: baseline operation under normal conditions, optimization with predictive restocking and promotional mechanisms, and failure conditions involving connectivity disruptions and synchronization delays.

8 Implementation & Results

8.1 Analysis Results

The data collection phase produced four primary datasets addressing distinct dimensions of the small business ecosystem.

Foot Traffic Comparison revealed that 343 entries were recorded on Saturday morning compared to 811 on Sunday afternoon — more than double — indicating that Sunday represents the peak traffic opportunity. However, exit counts suggest that a large proportion of Sunday visitors do not complete purchase transactions, pointing to a low conversion rate despite high foot traffic.

Visit Motives Distribution showed that only 11.76% of surveyed visitors came to the mall primarily to shop at small or local stores. Food court and restaurant visits (17.65%), cinema and entertainment (17.65%), and other motives (29.41%) accounted for the majority of traffic.

Post-Pandemic Spending Behavior indicated that 41.18% of respondents reported spending less than before the pandemic, while 29.41% reported spending more and 29.41% the same.

Inventory Rotation Frequency showed that 80% of surveyed store owners reported fast or very fast inventory rotation within two weeks, with no stores reporting slow rotation, indicating a high stockout risk under demand surges.

Table 1: Foot Traffic Comparison

Day	Entries	Exits
Saturday (9am–12pm)	343	262
Sunday (12pm–3pm)	811	309
Total	1,154	571

Table 2: Visit Motives Distribution

Motive	Frequency	Percentage
Shop in small / local stores	2	11.76%
Food court / Restaurants	3	17.65%
Cinema / Entertainment	3	17.65%
Large stores	2	11.76%
Just for a walk / Accompanying	2	11.76%
Other	5	29.41%
Total	17	100%

Table 3: Post-Pandemic Spending Behavior

Behavior	Frequency	Percentage
Greater than before	5	29.41%
Same as before	5	29.41%
Smaller than before	7	41.18%
Total	17	100%

Table 4: Inventory Rotation Frequency

Rotation Speed	Frequency	Percentage
Very Fast (less than a week)	2	40%
Fast (1–2 weeks)	2	40%
Normal (2–4 weeks)	1	20%
Slow (more than a month)	0	0%
Total	5	100%

8.2 System Design

In response to the analytical findings, a three-layer analytical platform was designed. The **input layer** captures data from customers, business managers, and products, filtered through external factors including weather conditions, economic conditions, and government regulations. The **processing layer** integrates six core components: Stores, Employees, Products, Inventory Management, Data Analysis, and a Recommendation System, connected through a Reporting Module. The **output layer** delivers Customer Satisfaction indicators, Revenue metrics, and structured Reports for decision-making.

The technical architecture separates the system into a front-end layer (UI interfaces, forms, dashboards, and data queries), a back-end layer (API control, business logic, system validations, and security), and a database layer (storing users, business data, reports, CRUD operations, and logs).

8.3 Risk and Quality Assessment

The risk management plan identified three primary risk categories. Technical risks include connectivity instability in certain mall zones and data inconsistency resulting from

synchronization failures. Operational risks include the high inventory rotation rate that increases stockout exposure and the variability of foot traffic driven by external factors such as weather and day of the week. User-related risks include resistance from business owners to share operational data and adopt new technologies.

Mitigation strategies include offline synchronization mechanisms, data validation protocols, a gradual pilot-based deployment strategy, and dedicated owner training. Quality acceptance criteria were established as follows: 99.5% system uptime during operating hours, dashboard response time within acceptable ranges, inventory alert reliability validated against stockout events, and a minimum 70% positive usability feedback from pilot participants.

8.4 Simulation Results

Three scenarios were evaluated across 100 simulation days. Under **baseline conditions**, cumulative sales reached approximately 510 units with stable inventory cycles and response times fluctuating between 1.0 and 1.9. Synchronization operated without disruption.

Under **optimization conditions**, cumulative sales exceeded 1,200 units — more than double the baseline — driven by predictive restocking and promotional mechanisms. Response times decreased to a range of 0.6 to 1.5, and synchronization delays remained at zero throughout the simulation.

Under **failure conditions**, cumulative sales dropped to approximately 420 units — the lowest across all scenarios. Inventory exhibited erratic depletion with repeated near-zero levels, and synchronization delays fluctuated between 0 and 8 units, directly degrading operational responsiveness.

Table 5: Simulation Results Summary

Metric	Baseline	Optimization	Failure
Cumulative Sales (100 days)	≈510	≈1,200	≈420
Avg. Response Time	1.0–1.9	0.6–1.5	N/A
Max Sync. Delay	0	0	8
Inventory Stability	Stable	Dynamic	Erratic

An emergent reinforcing feedback loop was identified in the optimization scenario: predictive restocking and active promotions combined to amplify sales growth beyond what either mechanism would generate independently. In the failure scenario, a cascading degradation pattern emerged where connectivity disruptions triggered synchronization failures, which delayed inventory alerts, which led to stockout events, revealing a critical interdependency between system components.

9 Discussion

The analysis of Titan Plaza confirms that the primary challenge facing small businesses is not the absence of customers, but the absence of information infrastructure that allows

merchants to engage those customers effectively. Large anchor stores circumvent this problem through digital channels that allow customers to verify product availability before visiting. Without equivalent tools, small merchants remain dependent on accidental walk-in traffic and informal word-of-mouth — both highly sensitive to external variables.

The simulation results directly validated this finding. The persistent gap between potential and actual buyers observed across all scenarios confirmed that foot traffic alone is insufficient to drive sales, and that the system’s active engagement features — promotions, alerts, and recommendations — are essential components rather than optional enhancements.

The failure scenario revealed synchronization reliability as the system’s most critical vulnerability. Relatively small connectivity disruptions produced disproportionate cascading effects across inventory management, sales processing, and operational responsiveness, elevating offline synchronization from a secondary feature to a core deployment priority.

The optimization scenario demonstrated that the combination of predictive mechanisms produces emergent amplification effects, confirming that the modular architecture enables component interactions that improve overall system performance beyond the sum of individual features.

Owner reluctance to share business data remains the most significant adoption barrier, and the gradual pilot-based deployment strategy was specifically designed to address this through trust-building before full rollout. The design prioritizes practical feasibility over technical sophistication, reflecting the reality that a highly advanced system that owners do not use delivers no value.

10 Conclusion

This report documents a complete systems engineering study of an analytical support platform for small businesses in Titan Plaza, progressing through four structured phases: analysis, design, robust engineering and management, and computational simulation.

The central finding confirmed across all phases is that the main limitation faced by small businesses is not the absence of customers, but the absence of tools to engage them effectively. The proposed platform addresses this through a modular three-layer architecture that centralizes data, automates critical operational alerts, and provides store owners with decision-making tools previously available only to large retail chains.

Simulation results validated the architectural decisions, demonstrating that optimization mechanisms more than doubled baseline sales performance while maintaining operational stability. The failure scenario confirmed that synchronization reliability is the system’s most critical operational dependency.

Future development should focus on deploying the system in a real environment, measuring impact through sales performance and customer satisfaction indicators, integrating loyalty programs and customer-facing features, and expanding the multi-mall model to generate cross-location benchmarks that further strengthen the competitive position of small businesses across Bogotá’s retail landscape.

References

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A Primary Data Tables

All primary data tables are included in Section 8.1 of this report.

B Simulation Scenario Graphs

Simulation output graphs for all three scenarios (Baseline, Optimization, and Failure) are documented in Workshop #4 and should be inserted as figures in this section using the corresponding image files.

Glossary

Conversion Rate: The proportion of mall visitors who complete a purchase transaction at a small business relative to total foot traffic.

Foot Traffic: The number of people entering and moving through a physical retail space within a given time period.

Informational Disconnect: The structural gap between the data available to large retail chains and the data available to small independent businesses, resulting in unequal capacity to respond to customer demand.

Inventory Rotation: The rate at which a store's stock is sold and replenished over a given period.

Offline Synchronization: A mechanism that allows a system to continue operating under limited or no internet connectivity, storing data locally and syncing with a central server when connection is restored.

Dashboard: A visual interface that aggregates and displays key performance metrics in real time, enabling informed decision-making by store owners.

Recommendation System: An algorithmic component that suggests products or actions based on historical data, behavioral patterns, and demand trends.

Emergent Behavior: System-level patterns or outcomes that arise from the interaction of individual components and cannot be predicted by analyzing those components in isolation.

Simulation: A computational model that replicates real-world system behavior to test design assumptions and evaluate performance under different conditions.

Word-of-Mouth: Informal peer-to-peer communication about a business or product, identified as a primary driver of customer acquisition for small businesses.